

Dr. Thejus Mahajan
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thejus.mahajan@uni-hamburg.de

23 January 2026

Prof. Dr. Hubert Preissl
Institute of Diabetes Research and Metabolic Diseases (IDM)
Helmholtz Munich / German Center for Diabetes Research
Otfried Müller Straße 10
72076 Tübingen

Application: Bioinformatician / Data Scientist (Job ID: 102834)

Dear Prof. Dr. Preissl,

I am writing to express my strong interest in the Bioinformatician / Data Scientist position at the Institute of Diabetes Research and Metabolic Diseases (IDM). The focus on statistical analysis of clinical data and the characterization of sub-phenotypes in prediabetes represents exactly the kind of impactful health research I am seeking to contribute to.

After completing my Ph.D. in astrochemistry and a postdoctoral position in computational modeling, I made a deliberate decision to redirect my career toward clinical bioinformatics. This was not a step backward, but a purposeful transition toward research that directly impacts human health. To achieve this, I enrolled in intensive training in bioinformatics and biostatistics at CQ Beratung + Bildung in Berlin, where I am acquiring expertise in NGS analysis, applied biostatistics, and clinical data analysis using R (Bioconductor) and Python. Crucially, this training includes hands-on work with real and simulated patient datasets—ANOVA, PCA, hierarchical clustering, and survival analysis in clinical contexts.

Additionally, I will be completing an internship at HealthTwist GmbH (a health technology company) focused on implementing Machine Learning workflows using the Tidymodels framework in R. This experience is directly relevant to developing predictive models for clinical research, such as identifying sub-phenotypes in prediabetes.

I bring a strong foundation in statistical analysis and computational modeling:

- During my postdoctoral work, I developed the “Cyanobacteria Life Cycle” model, analyzing large datasets using Python and R, and performing multivariate statistical analyses—skills directly applicable to organ crosstalk and clinical data analysis.
- My Ph.D. involved processing terabytes of experimental data, developing algorithms, and building predictive models—demonstrating my ability to handle complex, high-dimensional data.

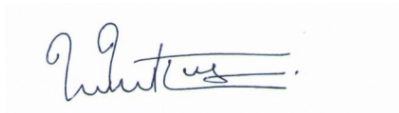
- I am proficient in Python (Pandas, NumPy, Scikit-learn), R (Bioconductor, Tidymodels, ggplot2), SQL, and pipeline development.

My motivation for this transition is genuine: I want to apply my computational skills to research that directly benefits patients. The work at IDM on prediabetes prevention and personalized therapy aligns perfectly with this goal. While this position requires a BSc-level qualification, I see this as an opportunity to establish myself in clinical bioinformatics and grow within this field. I am committed to contributing meaningfully and developing deep expertise in this domain.

I am particularly excited about the opportunity to develop analysis pipelines for the research groups and to support scientific publications and grant writing—areas where my experience with five peer-reviewed publications and academic research workflow will be valuable.

I have attached my CV and certificates for your review. I would welcome the opportunity to discuss my qualifications in an interview and am available at your convenience.

Hamburg, 23 January 2026

A handwritten signature in blue ink, appearing to read 'Thejus Mahajan', is displayed on a light blue rectangular background.

Dr. Thejus Mahajan

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Ph: +49-15259697629
Website: <https://thejusmahajan.github.io>
Hamburg, Germany

CAREER INTERESTS To apply expertise in computational modeling, data science, and bioinformatics to address complex challenges in clinical and biological research. Skilled in multivariate statistical modeling, biostatistics, and developing analysis pipelines using R (Bioconductor, Tidymodels) and Python. Actively transitioning into clinical bioinformatics through specialized training on patient datasets and biostatistical methods for health research.

- WORK EXPERIENCE
- **CQ Beratung + Bildung, Berlin, Germany**
Further Training – Application-related Bioinformatics and Biostatistics
August 2025 – Present
 - Shell (BASH), Python (object-oriented), and programming methodology.
 - Bioinformatics databases (NCBI, Biopython, SQL).
 - Sequence analysis, structural bioinformatics, and NGS analysis (Nextflow, Galaxy).
 - Applied biostatistics with real and simulated patient datasets (R/Bioconductor, ANOVA, PCA, HCA, survival analysis).
 - **HealthTwist GmbH, Berlin, Germany**
Internship – Machine Learning in R
February 2026 – April 2026
 - Implementation of Machine Learning workflows using the Tidymodels framework in R.
 - Supervised by Dr. Andreas Busjahn (HealthTwist GmbH).
 - Practical application of bioinformatics and biostatistical analysis skills.
 - **Helmholtz-Zentrum Hereon, Geesthacht, Germany**
Guest Scientist – Computational Biologist & Systems Modeler
May 2025 – October 2025
 - Spearheaded research modeling evolutionary processes in marine environments.
 - Simulated long-term impacts of environmental changes on ecosystems.
 - **University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences**
Post-doctoral Scientist – Marine Ecosystem Modelling
August 2021 – January 2025
 - Developed the “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework.
 - Engineered models in Fortran and analyzed large NetCDF datasets using Python in Linux.
 - Statistical analysis and visualization of ecological time-series data using R.

	• Université Paris-Saclay, Institut des Sciences Moléculaires d'Orsay, France Ph.D. Researcher in Astrochemistry October 2015 – September 2018 - Led atomic collision experiments and processed terabytes of raw data. - Utilized C++ for signal processing and algorithm development. - Developed a molecular fragmentation model for the KIDA database. - Thesis: “Excitation and fragmentation of C_nN^+ ($n=1-3$) molecules in collisions with He atoms at intermediate velocity; fundamental aspects and application to astrochemistry.” - Advisor: Dr. Karine Béroff	
EDUCATION	• Ph.D. in Astrochemistry (2015 – 2018) Institut des Sciences Moléculaires (ISMO), Université Paris-Saclay, France • M.Sc. in Physics (July 2012 – January 2015) National Institute of Technology Calicut, India CGPA: 8.2/10	
SKILLS & COMPETENCIES	Programming: Bioinformatics: Biostatistics: Data Analysis: Tools:	Python (Pandas, NumPy, Scikit-learn), R (Bioconductor, Tidy-models, ggplot2), Fortran, SQL, BASH NGS Analysis, Galaxy, Sequence Alignment, Structural Bioinformatics, Nextflow Multivariate Analysis, PCA, ANOVA, Dimensionality Reduction, Markov Chains Large dataset processing (NetCDF, HDF5), Data visualization, Pipeline development Linux/Unix, Git, HPC clusters, $\text{\LaTeX}$, Microsoft Office
LANGUAGES	English: German: French: Malayalam: Tamil, Hindi:	Proficient (speaking, reading, writing) B1 (Goethe Certified) Intermediate Native Proficient
TRAINING & COURSES	- HPC Training at Jülich Supercomputing Center (JSC) - MITx: 7.00x Introduction to Biology – The Secret of Life	
PUBLICATIONS	Google Scholar: https://scholar.google.com/citations?hl=en&user=PJkZwAwAAAAJ 1. T. Mahajan, et al. <i>Journal of Physics: Conference Series</i>, 1412:142026, 2020. 2. T. Idbarkach, et al. <i>Astronomy & Astrophysics</i>, 628:A75, 2019. 3. T. Mahajan, et al. <i>Journal of Physics B</i>, 2019. 4. T. Idbarkach, et al. <i>Journal of Physics B</i>, 51(24):245201, 2018. 5. T. Idbarkach, et al. <i>Journal of Physics: Conference Series</i>, 1412(11):112028, 2020.	

REFERENCES

- **Dr. Andreas Busjahn**
HealthTwist GmbH, Berlin
Email: andreas@busjahn.net
Professional relation: Biostatistics instructor and internship supervisor
- **Prof. Dr. Kai Wirtz**
Helmholtz-Zentrum Hereon, Ökosystemmodellierung
Email: kai.wirtz@hereon.de
Professional relation: Supervisor during Guest Scientist position
- **Prof. Dr. Elisa Schaum**
University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences
Email: elisa.schaum@uni-hamburg.de
Professional relation: Collaborator during Post-doctoral research

Deposited under the number : 2018SACLS311

CERTIFICATE OF ACHIEVEMENT

The president of Université Paris-Saclay certifies that the doctoral degree from Université Paris-Saclay, prepared at Université Paris-Sud, within the doctoral school n°572 - School EDOM Waves and Matter, in the specialty lasers, molecules, atmospheric irradiation, was awarded to

Monsieur THEJUS MAHAJAN

Born on September 30th 1992 at THRISSUR, INDE

Under the academic year 2018-2019

Title of the dissertation : Excitation and fragmentation of C_nN^+ ($n=1-3$) molecules in collisions with He atoms at intermediate velocity ; fundamental aspects and application to astrochemistry

Date of the defense : September 28th 2018

Place of the defense : Amphithéâtre du bât. 520 (3ème étage), ISMO Institut des Sciences Moléculaires d'Orsay - UMR 8214 Rue André Rivière Bâtiment 520 Université Paris-Sud 91405 Orsay Cedex

Composition of the Jury

M. Alain DUBOIS, Professeur, Université Pierre-et-Marie-Curie - President

M. Patrick ROUSSEAU, Maître de Conférences, UNIVERSITÉ DE CAEN NORMANDIE - Reviewer

M. Jean-Christophe LOISON, Directeur de Recherche, Institut des Sciences Moléculaires de Bordeaux - Reviewer

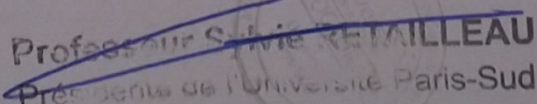
Mme Séverine BOYÉ-PÉRONNE, Professeure, Université Paris-Saclay - Examiner

Mme Karine BEROFF, Directeur de Recherche, CNRS et Université Paris-Sud - Thesis Supervisor

Paris-Saclay

October 26th 2018

The president of Université Paris-Saclay or his delegate


Professeur Sylvie RETAILLEAU
Président de l'Université Paris-Sud

RÉPUBLIQUE FRANÇAISE

MINISTÈRE DE L'ÉDUCATION NATIONALE, DE L'ENSEIGNEMENT SUPÉRIEUR ET DE LA RECHERCHE

université
PARIS-SACLAY

DOCTORAT

Vu le code de l'éducation, notamment ses articles L.612-7, L. 613-1, D. 613-3 et D. 613-6 ;

Vu le code de la recherche, notamment son article L.412-1 ;

Vu l'arrêté du 25 mai 2016 fixant le cadre national de la formation et les modalités conduisant à la délivrance du diplôme national de doctorat ;

Vu l'arrêté du 10 juillet 2015 portant accréditation de l'école doctorale ;

Vu les pièces justificatives produites par Monsieur Thejus MAHAJAN, né le 30 septembre 1992, à Thirissur (Inde) en vue de son inscription en doctorat ;

Vu la procès-verbal du jury attestant que l'intéressé a soutenu, le 28 septembre 2018, une thèse portant sur le sujet suivant : **Excitation et fragmentation des molécules C₆H⁺ (n = 1-3) en collision avec des atomes de He à vitesse intermédiaire: aspects fondamentaux et application à l'astrochimie préparée au sein de l'école doctorale Ondes et Matière** devant un jury présidé par M. Alain DUBOIS et composé de Mme Karine BEROFF, M. Patrick ROUSSEAU, M. Jean Christophe LOISON, Mme Séverine BOYÉ-PÉRONNE ;

Vu la délibération du jury :

Le diplôme de **DOCTORAT** en **lasers, molécules, rayonnement atmosphérique préparé à l'université Paris-Sud** est délivré à Monsieur **Thejus MAHAJAN** au titre de l'année universitaire 2018-2019 et confère le **grade de docteur**, pour en jouir avec les droits et prérogatives qui y sont attachés.

Fait le 05 février 2019,

Le titulaire

La Présidente
de l'université Paris-Saclay

La Rectrice d'Académie,
Chancellerie des universités

Numéro du diplôme : 2018SACLS3112019036000005446

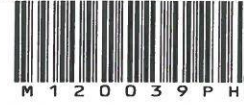
SYME RETAILLEAU

Chantime AVENEL



No.: $\frac{A}{14}$ 01819

राष्ट्रीय प्रौद्योगिकी संस्थान कालिकट
National Institute of Technology Calicut



Upon the recommendation of the Senate hereby confers on

THEJUS MAHAJAN

the degree of

*Master of Science
in Physics*

in the year 2014

and places him/her in First Class with

Distinction

Given at the National Institute of Technology Calicut

on 10th January, 2015


Registrar


Director

യൂണിവേഴ്സിറ്റി ഓഫ് കാലിക്കറ്റ്



UNIVERSITY OF CALICUT
FACULTY OF SCIENCE

Whereas it has been certified by the duly appointed Examiners that

Thejus Mahajan

is qualified to receive the Degree of Bachelor of Science (B.Sc.) in

Physics

he having successfully completed the programme in April 2012

with Register No. SKAJSPH033

and been placed in overall grade B+

with Complementary Courses *Mathematics* and *Chemistry*

The Senate of the University of Calicut hereby confers on him

the Degree of

Bachelor of Science

with all the Rights, Privileges and Honours thereunto appertaining.

Given under the seal of the University

Cellusana

Vice Chancellor

CALICUT UNIVERSITY P.O.

Kerala, INDIA

PIN - 673 635

Date..22/02/2014.....

Digitally signed by DR. M ABDUL SALAM

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